

AZ-104 Lab 03 – Manage Azure Resources using ARM Templates & Bicep

Lab Objective

This lab introduces Infrastructure as Code (IaC) in Azure using Azure Resource Manager (ARM) templates and Azure Bicep. The goal is to automate deployments, reduce human error, and ensure consistent resource provisioning across environments.

Why Infrastructure as Code?

Manual deployments through the Azure Portal are error-prone and inconsistent. Infrastructure as Code allows teams to define infrastructure declaratively, store it in version control, and deploy it reliably using automation tools.

Core Concepts

- 1 Azure Resource Manager (ARM): The central deployment and management service for Azure.
- 2 ARM Templates: JSON files that declaratively define Azure infrastructure.
- 3 Parameters and Variables: Enable template reuse and flexibility.
- 4 Azure Cloud Shell: Browser-based environment for PowerShell and CLI.
- 5 Azure Bicep: A simplified DSL that compiles to ARM JSON.

ARM Template Structure

ARM templates follow a defined JSON structure. Each section plays a specific role in describing the desired state of Azure resources.

- 1 `$schema` – Defines the JSON schema used for validation.
- 2 `contentVersion` – Versioning for the template.
- 3 `parameters` – Inputs provided at deployment time.
- 4 `variables` – Reusable internal values.
- 5 `resources` – The Azure resources to deploy.
- 6 `outputs` – Values returned after deployment.

Lab Tasks Explained

- 1 Task 1: Create a managed disk and export its ARM template.
- 2 Task 2: Modify the ARM template and redeploy it.
- 3 Task 3: Deploy ARM templates using Azure PowerShell.
- 4 Task 4: Deploy ARM templates using Azure CLI.
- 5 Task 5: Deploy a managed disk using Azure Bicep.

ARM Templates vs Azure Bicep

- 1 ARM templates use verbose JSON syntax.
- 2 Bicep uses concise, readable syntax.
- 3 Bicep compiles into ARM templates.
- 4 ARM remains the underlying deployment engine.

Interview-Ready Questions and Answers

Q: What is Azure Resource Manager?

A: Azure Resource Manager is the deployment and management layer that enables consistent provisioning and management of Azure resources.

Q: What is an ARM template?

A: An ARM template is a JSON file that declaratively defines Azure infrastructure.

Q: Why are parameters used in ARM templates?

A: Parameters allow the same template to be reused across different environments.

Q: How does Azure Bicep differ from ARM templates?

A: Azure Bicep is a higher-level language that simplifies ARM template authoring and compiles into ARM JSON.

Q: How can ARM templates be deployed?

A: ARM templates can be deployed using the Azure Portal, PowerShell, CLI, or CI/CD pipelines.

Q: Why is Infrastructure as Code important?

A: Infrastructure as Code enables automated, repeatable, and consistent deployments.

Key Takeaways

ARM templates and Bicep enable Infrastructure as Code in Azure. Templates can be deployed through multiple tools. Bicep simplifies authoring while using ARM internally. IaC is foundational for DevOps and cloud automation.